MILITARY SPECIFICATION
REMOVER, AIRCRAFT CANOPY: M9
REMANUFACTURED METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Remover, Aircraft Canopy: M9 Remanufactured Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-C-5541	- Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloy

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1168	- Ammunition Data Card

DRAWINGS

Munitions Command

FF9878	- Remover, Aircraft Canopy: M9 Metal Parts Assembly
C8846598	- Box, Packing Ammunition, for Remover, Aircraft Canopy: M9 with Cartridge, M151

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2.1 DRAWINGS (Cont'd)

Munitions Command
C8846599 - Box, Inner, Packing, Ammunition for Remover, Aircraft Canopy: M9 with Cartridge, M151

IEL-9878 - Inspection Equipment List for Remover, Aircraft Canopy: M9, Metal Parts Assembly

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Hydrostatic pressure.- The components listed in Table I shall withstand a constant internal pressure as indicated in Table I for a period of 15 seconds, minimum without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.1).

TABLE I

Component	Hydrostatic Pressure (psi, min)
Tube, Ballistic Chamber	3000
Tube, Trunnion	3000
Tube, Outside	2200
Tube, Ballistic Assembly	3000

3.3 Manual function.- Without electrical power and with the clutch disengaged, the pull force required to extend the inside tube assembly shall not exceed 40 pounds.

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3.4 Reversibility load.- The remover shall be capable of withstanding a load of 750 pounds minimum in compression, and 950 pounds minimum in tension, applied to the remover mounting points. The remover shall not travel due to the applied loads after being stopped at any part of its stroke (See 6.1).

3.5 Clutch disengage load.- The force, applied at the cable assembly, required to accomplish clutch disengagement shall not exceed 20 pounds.

3.6 Electrical operation.- The remover shall be cycled electrically to the load curves A, B, and C of Figure 1. The remover shall be capable of operating to the loads in Figure 1 and shall comply with the following requirements (See 6.2).

3.6.1 There shall be no structural failure of any part under the inertia loads imposed by the sudden interruption of electric power.

3.6.2 Time required for linear extension or retraction of the remover for a normal stroke of 45.62 inches and under any load defined in the load-stroke curve of Figure 1, shall be 23 seconds maximum and 10 seconds minimum measured from initial application of power. Voltage input shall be 28 Volts, D.C.

3.6.3 The motor current drain under any load defined in the load-stroke curve of Figure 1, shall not exceed 26 amperes.

3.6.4 The load limit switch which de-energizes the motor at completion of the retracted stroke, shall operate in a range equivalent to the remover load (cinch-down load) of 400 to 500 pounds. The cinch-down load shall be checked at completion of duty cycle when performing electrical operation under normal load, Curve B, Figure 1.

3.7 Breakaway torque.- The assembled remover shall withstand the following breakaway torque without movement or distortion:

TABLE II

Component	Breakaway Torque (inch pounds)
Tube, Trunnion to Adapter, Housing	400 min.
Tube, Outside to Tube, Trunnion	400 min.
Tube, Ballistic Chamber to Adapter, Housing	400 min.

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3.8 Metal parts assembly.- Parts shall be combined to produce a metal parts assembly which conforms to the metal parts assembly and the following:

3.8.1 The remanufactured remover metal parts assembly shall contain new parts in accordance with Table III:

TABLE III

Name of Part		Drawing Number
Actuator Switch		FA29924
Actuator Switch		FA29928
Cap, Plastic Dust		MS25177-16
Clamp, Cushion Support - 3 req'd		MS21919-H5
Clamp, Cushion Support		MS21919-H8
Detent Assembly		FB51687
Gasket, Support Work Shaft		B10533849
Gasket, Cover, Helical, Gear		B10533850
Gasket, Cover Worm		B10543073
Grommet, Elastic - 2 req'd	AN931B3-5 *	(MS35489-62) **
"O" Ring	CKCX2-501220 *	(MS28775-012) **
"O" Ring		MS28775-013
"O" Ring	CKCX2-501225 *	(MS28775-116) **
"O" Ring	CKCX2-501236 *	(MS28775-222) **
"O" Ring		MS28775-332
"O" Ring		MS29513-26
"O" Ring - 2 req'd		MS29513-130
"O" Ring		MS29513-142
"O" Ring		MS29513-146
"O" Ring		MS29513-147
Pin, Cotter Split		MS24665-5
Plug, Shipping		A8593299
Screw, Shear - 2 req'd		FB52094
Seal		A10533925
Seal, Oil		A10533851
Shipping Lock Assembly		FB51605
Sleeve (Bulk Material)		FA29965
Spring		FA31205
Spring		FB52070
Spring, Clutch		FB51963
Spring, Helical Torsion		FB51921

* Old Standard
** Latest Standard

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3.8.2 Any spring pin removed from an assembly shall be replaced with a new spring pin.

3.8.3 Any bolt, nut, screw insert or any other threaded component which shows evidence of cross-threading, stripping or excessive wear shall be replaced with a new part.

3.8.4 Bearing and ball bearing screw assembly.- Any bearing and ball bearing screw assembly which shows evidence of roughness, binding or damage shall be replaced with a new part.

3.8.5 Switch.- Any switch which contains a loose plunger, gummy, or damaged plunger seal or damaged wiring shall be replaced with a new switch.

3.8.6 D.C. electric motor.- Replace defective motor if minor repairs cannot be accomplished.

3.8.7 Gear.- Any gear with excessive backlash, worn or broken teeth, or improper mating shall be replaced with a new gear.

3.8.8 Gear box and ballistic assembly.- Any part in the gear box and ballistic assembly which shows evidence of distortion, excessive wear, looseness or any other damage shall either be repaired or replaced with a new part.

3.8.9 Gear box lubrication.- The grease in the gear box cavity (worm gear and helical gear cavities) shall be removed and replaced with fresh grease conforming to the requirements of the applicable grease specification.

3.8.10 The housing sub-assembly shall be modified in accordance with instructions contained in 6.4.

3.9 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied (see 6.3).

3.10 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.10.1 Metal defects.- The part shall be free from cracks, splits, or other metal defects.

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3.10.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.3 Electrical wire and connections.- The electrical wire shall be not frayed, cut or broken and wire connection shall be not loose or broken.

3.10.4 Foreign matter.- No part or assembly shall contain chips, dirt, grease (except where required) rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the Quality Assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 Removers, maximum, exclusive of test samples, provided the homogeneity of remanufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

4.2.2 Examination.- The roughness comparison specimens prescribed by ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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4.2.2.1 Reclaimed parts.- Examination of reusable parts, which require no reprocessing shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be rejected.

4.2.2.2 Reprocessed parts.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defects within a major or minor group, the criteria in Table V shall be used. All nonconforming material shall be rejected.

TABLE IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE V

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	7	7
20 - 43	4	5	8	10
Over 43	6	8	11	14

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4.2.2.2 (Cont'd)

TABLE V (Cont'd)

Sample Sizes for Minor Defects	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.2.2.3 Housing, Rod End, Drawing FB51849 covering a detail of Drawing FF9878

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Maximum diameter of thread, min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Rod End, Drawing FB51851 covering a detail of Drawing FF9878

CRITICAL

None defined

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4.2.2.4 (Cont'd)

Defect		Method of Inspection
MAJOR		
101.	Pitch diameter of thread, max.	Gage
102.	Minor diameter of thread, max.	Gage
103.	Diameter of bore, max.	Gage
104.	Metal defective	Visual
105.	Protective finish defective or inadequate	Visual
106.	Burr	Visual
107.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.5 Cover, Worm, Drawing FB51862 covering a detail of Drawing FF9878

CRITICAL		
	None defined	
MAJOR		
101.	Metal defective	Visual
102.	Protective finish defective or inadequate	Visual
103.	Burr	Visual
104.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.6 Cover, Helical Gear, Drawing FF51864 covering a detail of Drawing FF9878

CRITICAL	
None defined	

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4.2.2.6 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Metal defective	Visual
102. Protective finish defective or inadequate	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.7 Cover, Side, Drawing FB51868 covering a detail of Drawing FF9878

CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish defective or inadequate	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.8 Bracket, Clutch Release, Cable, Drawing FB51890 covering a detail of Drawing FF9878

CRITICAL	
None defined	

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4.2.2.8 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Metal defective	Visual
102. Protective finish defective or inadequate	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.9 Port, Gas, Drawing FB51891 covering a detail of Drawing FF9878

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Metal defective	Visual
104. Protective finish defective or inadequate	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
None defined	

4.2.2.10 Tube, Ballistic Chamber Drawing FB51892 covering a detail of Drawing FF9878

CRITICAL	
None defined	

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4.2.2.10 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Pitch diameter of large thread, min.	Gage
102. Maximum diameter of large thread, min.	Gage
103. Pitch diameter of small thread, min.	Gage
104. Maximum diameter of small thread, min.	Gage
105. Metal defective	Visual
106. Protective finish defective or inadequate	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
None defined	

4.2.2.11 Cam, Drawing FB51920 covering a detail of Drawing FF9878

CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish defective or inadequate	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.12 Ring, Drawing FB51925 covering a detail of Drawing FF9878

CRITICAL	
None defined	

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4.2.2.12 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Metal defective	Visual
102. Protective finish defective or inadequate	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.13 Plug, Drawing FB52069 covering a detail of Drawing FF9878

CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish defective or inadequate	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.14 Tube, Trunnion Drawing FC9589 covering a detail of Drawing FF9878

CRITICAL	
None defined	

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4.2.2.14 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Pitch diameter of inside thread, min.	Gage
102. Minor diameter of inside thread, min.	Gage
103. Pitch diameter of outside thread, max.	Gage
104. Maximum diameter of outside thread, max.	Gage
105. Metal defective	Visual
106. Protective finish defective or inadequate	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
None defined	

4.2.2.15 Tube, Extension Drawing FC9897 covering a detail of Drawing FF9878

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of small thread, min.	Gage
102. Maximum diameter of small thread, min.	Gage
103. Pitch diameter of large thread, min.	Gage
104. Maximum diameter of large thread, min.	Gage
105. Metal defective	Visual
106. Protective finish defective or inadequate	Visual
107. Coating defective or inadequate	Visual
108. Burr	Visual
109. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.16 Cap, Ballistic Chamber, Drawing FC9596 covering a detail of Drawing FF9878

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Metal defective	Visual
104. Protective finish defective or inadequate	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
None defined	

4.2.2.17 Housing Sub-Assembly, Drawing FD20805 covering a detail of Drawing FF9878

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Any part missing	Visual
102. Diameter of drilled holes (2) incorrect (modification)	Gage
103. Counterbore in drilled holes (2) incorrect (modification)	Gage
104. Holes not plugged	Visual
105. Rivets not flush	Visual
106. Metal defective	Visual
107. Protective finish defective or inadequate	Visual
108. Burr	Visual
109. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.18 Tube, Outside Drawing FD20806 covering a detail of Drawing FF9878

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of thread, min.	Gage
102.	Maximum diameter of thread, min.	Gage
103.	Pitch diameter of tapped holes, max.	Gage
104.	Minor diameter of tapped holes, max.	Gage
105.	Metal defective	Visual
106.	Protective finish defective or inadequate	Visual
107.	Burr	Visual
108.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.19 Adapter Housing, Drawing FD20807 covering a detail of Drawing FF9878

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of small inside thread, max.	Gage
102.	Minor diameter of small inside thread, max.	Gage
103.	Pitch diameter of large inside thread, max.	Gage
104.	Minor diameter of large inside thread, max.	Gage
105.	Pitch diameter of tapped holes, max.	Gage
106.	Minor diameter of tapped holes, max.	Gage
107.	Metal defective	Visual
108.	Protective finish defective or inadequate	Visual
109.	Burr	Visual
110.	Foreign matter	Visual
MINOR		
	None defined	

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4.2.2.20 Tube, Ballistic Assembly, Drawing FD20809 covering a detail of Drawing FF9878

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish defective or inadequate	Visual
103. Coating defective or inadequate	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.21 Metal Parts Assembly, Drawing FF9878

CRITICAL	
1. Length of firing pin striker	Gage
2. Improper or insecure assembly	Manual and Visual
3. Loose or improper wiring	Manual and Visual
4. Improper soldering	Visual
5. Gas port hole obstructed	Visual
6. Any part missing	Visual
7. Marking missing, incorrect or illegible	Visual
8. Burr	Visual
9. Foreign matter	Visual
MAJOR	
None defined	
MINOR	
None defined	

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4.2.2.22 Packing in Inner Box, Drawing C8846599

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual
202.	Top of box loose	Manual
203.	Strapping missing, loose or improperly assembled	Manual and Visual
204.	Supports or fillers missing from box	Visual
205.	Box damaged	Visual
206.	Seal missing or improperly positioned	Visual
207.	Motor wires loose	Visual
208.	Marking missing, incorrect or illegible	Visual

4.2.2.23 Packing in Outer Box, Drawing C8846598

CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual
202.	Liner, fillers, cushings or spacers missing	Visual
203.	Strapping missing, loose or improperly assembled	Visual
204.	Hardware missing, loose or improperly engaged	Visual
205.	Box damaged	Visual
206.	Contents of box exposed	Visual
207.	Car seal missing or improperly sealed	Visual
208.	Marking missing, incorrect or illegible	Visual

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4.2.3 Inspection.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Hydrostatic pressure.- Each component listed in Table I shall be subjected to the hydrostatic pressure test. Failure of any item to comply with the requirements of 3.2 shall be cause for rejection of the item.

4.2.3.2 Protective finish, replated components.- The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standards listed on the drawing.

4.2.3.3 Breakaway torque.- A breakaway torque shall be applied to each assembly threaded joint as listed in Table II. Failure of any assembly to comply with the requirements of Table II shall be cause for rejection of the assembly.

4.2.3.4 Manual function.- Each remover shall be tested for the pull force required to extend the inside tubes. Failure of any remover to comply with the requirements of 3.3 shall be cause for rejection of the remover.

4.2.3.5 Reversibility load.- Each remover shall be tested for reversibility load. Failure of any remover to comply with the requirement of 3.4 shall be cause for rejection of the remover.

4.2.3.6 Clutch disengage load.- Each remover shall be tested for the force required to accomplish clutch disengagement. Failure of any remover to comply with the requirement of 3.5 shall be cause for rejection of the remover.

4.2.3.7 Operating load (Curve B).- Each remover shall be cycled electrically 5 times to the load curve B of Figure I. Failure of any remover to comply with the requirements of 3.6 shall be cause for rejection of the remover.

4.2.3.7.1 Operating load (Curves A and C).- Each remover that has been tested and accepted for curve B shall be cycled electrically 5 times each to the load curves A and C of Figure I. Failure of any remover to comply with the requirements of 3.6.1, 3.6.2 and 3.6.3 shall be cause for rejection of the remover.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-9878.

4.3 Test methods and procedures.

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4.3.1 Hydrostatic pressure.- Prior to the application of protective finish, the components listed in Table I shall be mounted in the test device and subjected to an internal hydrostatic pressure to determine compliance with the requirements of 3.2. The test device used shall internally seal the tubes within $1\frac{1}{2}$ inches of each end without applying an axial load other than that applied by the liquid pressure.

4.3.2 Protective finish, replated components.- Test procedures for protective finish shall be in accordance with the applicable Military Standard referenced on the drawing.

4.3.3 Breakaway torque.- A torque wrench set at the torque requirement specified in Table II for the respective assembly or sub-assembly shall be used to determine the adequacy of the joint. The test shall be accomplished by securing the assembly or sub-assembly in a fixture.

4.3.4 Manual function.- The inside tube shall be extended manually with the clutch disengaged and without electric power.

4.3.5 Reversibility load.- The test shall be performed at zero inches, midpoint of stroke and at full stroke with the load in tension and compression as specified in 3.4.

4.3.6 Clutch disengage load test.- A force shall be applied along the longitudinal axis of the cable assembly using a fixture to mechanically record the applied load.

4.3.7 Operating loads.- The remover shall be secured in the fixture between remover mounting points and tested in accordance with 4.2.3.7 and 4.2.3.7.1. The indicated values obtained shall be recorded and used to determine compliance with 3.6.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking (Level A).- Packing and marking shall be in accordance with Drawings C8546596 and C8546509.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

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6.2 A complete operation cycle consists of clutch engagement, extension, retraction and cinch-down load application.

6.3 Remanufactured exterior components are listed in Table VI.

TABLE VI

Name of Part	Drawing Number
Housing, Rod End	FB51849
Rod End	FB51851
Cover, Worm	FB51862
Cover, Helical Gear	FB51864
Cover	FB51868
Bracket, Clutch Release Cable	FB51890
Port, Gas	FB51891
Tube, Ballistic Chamber	FB51892
Cam	FB51920
Ring	FB51925
Plug	FB52069
Tube, Trunnion	FC9589
Cap, Ballistic Chamber	FC9590
Tube, Extension (Emralon Coating)	FC9597
Housing Sub-Assembly (See 6.3.1)	FD20805
Tube, Outside	FD20806
Adapter Housing	FD20807
Tube, Ballistic Assembly (Emralon Coating)	FD20809

6.3.1 The Housing Sub-Assembly Casting and Cover Plate shall only be replated if obviously necessary.

6.3.2 Bare spots on electric motor shall be touched-up with black paint.

6.3.3 Exposure of base metal on all other internal aluminum components shall be treated with a chemical film, Grade A or B, Class 1, Specification MIL-C-5541.

6.3.4 Locking inserts.- Components which contain locking inserts and are to be replated shall have the locking inserts removed and new inserts installed, after plating, in accordance with the applicable drawing.

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6.4 Housing Sub-Assembly.- The cover plate on Housing Sub-Assembly FD20805 shall be modified to have the lower switch holes relocated as shown in Figure 2. The original holes shall be plugged in accordance with requirements of Figure 2.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A290

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M9 REMOVER
Load-Stroke Curves

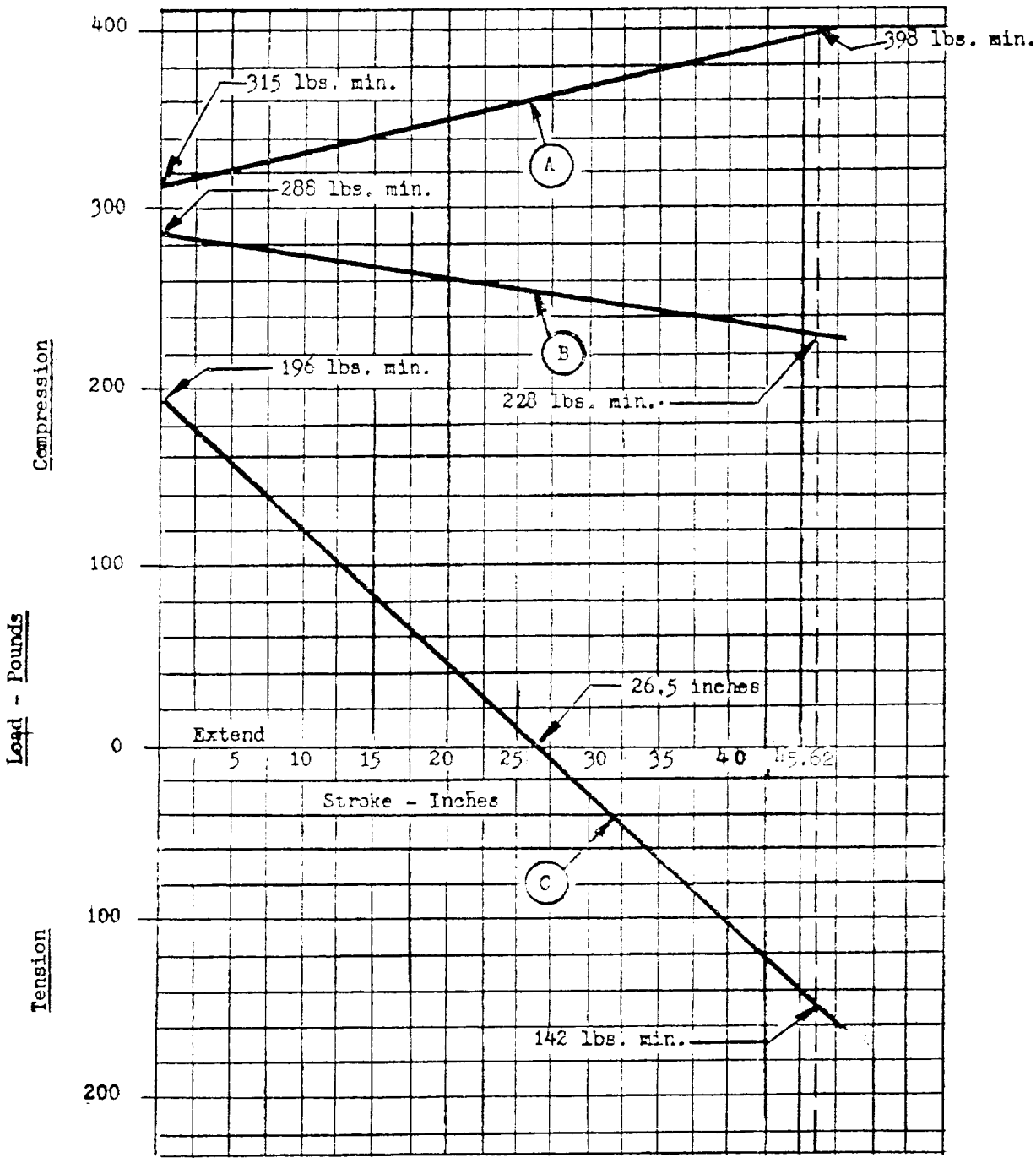
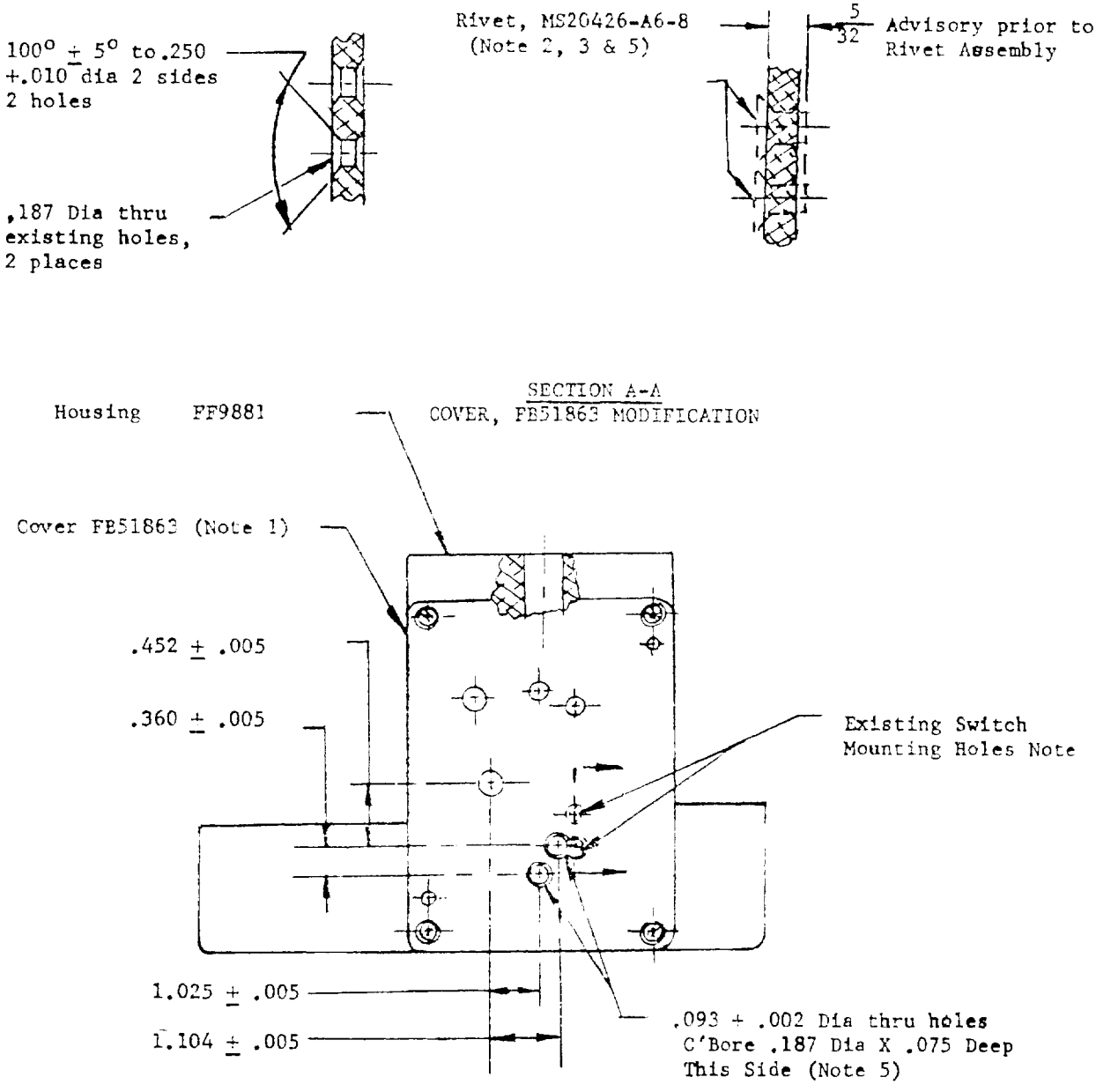


FIGURE 1

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FD 20805, HOUSING SUB-ASSEMBLY MODIFICATION OF



NOTE:
1- Modify cover, FB51863 in accordance with dimensions and requirements as shown herein.
2. Modify Rivets to length shown prior to assembly. Then squeeze both ends of each Rivet (Note 3).
3- Rivet material must be flush with both sides of cover.
4- Plug these holes before drilling relocated holes.
5- Apply a coat of chemical film, Spec MIL-C-5541 on all machined surfaces

FIGURE 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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NOTICE OF
VALIDATION

MIL-R-14821(OS)
NOTICE 1
15 March 1987

MILITARY SPECIFICATION

REMOVER, AIRCRAFT CANOPY: M9
REMANUFACTURED METAL PARTS ASSEMBLY

MIL-R-14821(MU) dated 24 July 1968, has been reviewed and determined to be valid for use in acquisition.

Preparing Activity:
Navy-OS

AMSC N/A

FSC 1377

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